

WORCESTER, MA, USA

WMO: 725100

Lat: 42.271N

Lon: 71.873W

Elev: 1000

StdP: 14.17

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94746

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.7	6.8	-12.7	2.9	5.2	-8.2	3.7	10.3	31.2	19.1	28.2	23.2	13.8	270	0.590

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.9	86.0	71.4	83.4	69.7	81.0	68.2	74.0	81.7	72.5	79.4	71.1	77.4	11.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.8	121.8	77.6	70.3	115.9	76.0	69.0	110.7	74.7	38.2	81.7	36.7	79.1	35.5	77.4	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.7	22.7	19.7	DB	-4.1	90.5	4.8	2.6	-7.5	92.4	-10.4	93.9	-13.1	95.3	-16.6	97.2	(4)
			WB	-5.2	77.1	4.6	2.2	-8.5	78.7	-11.3	79.9	-13.9	81.1	-17.2	82.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.5	24.7	27.2	34.6	46.4	56.6	65.4	70.9	69.4	62.5	51.0	40.4	30.8	
	DBStd	18.04	10.81	9.91	9.63	8.81	7.99	6.87	5.23	5.16	6.80	7.85	8.90	9.30	
	HDD50	3096	785	638	487	170	24	1	0	0	1	84	308	598	
	HDD65	6561	1249	1057	942	561	283	78	11	18	126	439	737	1060	
	CDD50	2533	1	1	11	62	229	463	649	603	376	114	21	3	
	CDD65	523	0	0	0	3	24	90	195	156	51	4	0	0	
	CDH74	3237	0	0	5	49	212	600	1259	856	244	12	0	0	
	CDH80	717	0	0	1	12	45	138	305	170	45	1	0	0	
Wind	WSAvg	10.1	11.5	11.6	11.4	10.8	9.8	9.1	8.8	8.3	8.6	9.9	10.5	11.1	
Precipitation	PrecAvg	47.0	3.5	3.3	4.1	3.9	4.1	3.9	3.8	3.8	4.2	4.0	4.3	3.9	
	PrecMax	71.7	11.2	8.4	8.0	8.8	9.9	12.2	7.9	8.0	13.1	10.2	10.4	9.8	
	PrecMin	32.0	0.9	0.2	0.7	0.7	1.2	0.3	0.7	1.0	0.7	0.7	0.7	0.7	
	PrecStd	7.8	2.2	1.5	1.7	1.9	1.9	2.6	1.8	1.7	2.5	2.0	2.0	2.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.2	56.8	70.3	81.1	84.9	88.0	89.7	88.0	85.2	75.3	65.8	60.7	
		MCWB	54.0	50.3	55.4	60.4	66.6	72.0	74.3	72.9	70.4	62.4	57.7	55.9	
	2%	DB	51.0	50.5	59.9	71.5	80.0	84.0	86.1	84.3	80.0	70.4	61.5	53.8	
		MCWB	47.7	44.5	48.8	55.3	63.2	70.0	72.1	71.2	67.7	62.1	56.0	50.0	
	5%	DB	44.8	45.4	53.6	65.6	75.1	80.5	83.4	81.5	76.4	66.7	57.6	49.2	
		MCWB	41.5	39.8	45.4	52.6	60.9	67.7	70.4	69.2	66.1	59.8	52.4	45.6	
	10%	DB	39.0	40.7	48.3	60.7	70.7	77.2	80.9	78.9	73.1	63.2	54.0	44.4	
		MCWB	35.5	35.8	41.9	49.6	58.8	65.2	69.0	67.6	64.5	57.0	48.9	40.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.4	53.6	57.3	63.3	69.6	74.6	76.5	75.2	73.1	68.0	62.0	56.4
			MCDB	56.4	56.0	68.2	73.8	81.0	83.8	85.4	83.4	78.8	70.8	63.9	58.0
2%		WB	48.4	46.2	52.2	59.1	66.5	72.3	74.4	73.4	70.8	64.6	57.7	50.9	
		MCDB	50.1	49.0	57.4	68.8	74.9	80.8	82.4	80.0	76.1	68.0	60.5	53.2	
5%		WB	41.5	40.5	46.9	55.4	64.0	70.4	72.8	71.9	69.0	61.2	53.9	46.0	
		MCDB	44.5	44.6	51.9	62.1	71.9	77.3	79.8	77.7	73.7	65.6	56.5	48.4	
10%		WB	35.8	36.2	42.2	51.4	61.0	68.2	71.3	70.4	67.0	57.7	49.8	40.9	
		MCDB	38.6	39.9	47.9	58.1	67.8	74.0	77.9	75.9	71.5	61.9	53.7	44.0	
Mean Daily Temperature Range	5% DB	MDBR	13.3	14.3	15.1	17.3	17.5	16.3	15.9	15.4	15.3	14.7	13.5	12.3	
		MCDBR	18.5	18.8	22.0	24.4	23.3	19.7	18.0	17.5	18.0	18.2	17.3	17.6	
		MCWBR	17.6	15.0	14.5	13.4	12.2	10.4	8.4	8.1	9.0	11.5	14.0	16.5	
	5% WB	MCDBR	18.8	18.0	20.6	22.6	21.1	18.0	16.5	15.5	15.4	16.7	16.8	17.4	
		MCWBR	18.3	16.1	16.0	15.1	12.6	10.9	8.7	8.4	9.8	13.4	15.7	17.6	
Clear-Sky Solar Irradiance	taub	0.278	0.288	0.315	0.359	0.394	0.433	0.450	0.420	0.359	0.322	0.297	0.280		
	taud	2.475	2.460	2.473	2.388	2.320	2.234	2.195	2.266	2.418	2.544	2.572	2.562		
	Ebn at Noon	279	293	297	289	279	267	261	266	277	278	271	270		
	Edh at Noon	23	28	31	36	40	44	45	41	32	25	21	20		
All-Sky Solar Radiation	RadAvg	575	847	1166	1486	1639	1779	1866	1676	1360	892	620	471		
	RadStd	39	71	81	159	186	175	110	85	98	84	51	49		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+164	N/A
Regional (79 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+171	+81

Nomenclature: See separate page